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ELECTRONIC APPARATUS AND HINGE DEVICE

Abstract

An electronic apparatus includes a first chassis, a second chassis, and a hinge device that connects the first chassis and the second chassis. The hinge device includes a first shaft fixed relative to the first chassis, a second shaft fixed relative to the second chassis, a support member configured to support the first shaft and the second shaft so as to be rotatable relative to each other with axial directions parallel to each other, a first slider having a first end portion facing a first shaft side and a second end portion facing a second shaft side and configured to be slidable between the first shaft and the second shaft, and a second slider having a first end portion facing the first shaft side and a second end portion facing the second shaft side and configured to be slidable between the first shaft and the second shaft.

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Background/Summary

BACKGROUND OF THE INVENTION

Cross-Reference to Related Applications

[0001] This application claims priority to Japanese Patent Application No. 2024-037314 filed on Mar. 11, 2024, the contents of which are hereby incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to an electronic apparatus including a hinge device that connects two chassis to be rotatable, and the hinge device.

DESCRIPTION OF THE RELATED ART

[0003] For example, Japanese Patent No. 5976052 discloses an electronic apparatus of a convertible type PC in which a second chassis having a display is connected to a first chassis having a keyboard to be rotatable in an angle range of 0 degrees to 360 degrees. In this configuration, a hinge device having a biaxial structure with two rotation shafts corresponding to each chassis is used.

SUMMARY OF THE INVENTION

[0004] In the hinge device of Japanese Patent No. 5976052, cam members having recessed portions are fixed to each of the two rotation shafts, and one floating pin is selectively fitted into each of the recessed portions. Accordingly, the hinge device is able to selectively rotate each of the rotation shafts according to an angle range between the chassis. Specifically, only the rotation shaft on the second chassis side having the display rotates from 0 degrees to 180 degrees, and only the rotation shaft on the first chassis side having the keyboard rotates from 180 degrees to 360 degrees. As a result, the shafts smoothly rotate within the angle between chassis from 0 degrees to 360 degrees.

[0005] As described above, the configuration of Japanese Patent No. 5976052 switches which of the two rotation shafts is rotated at a position of 180 degrees. Therefore, in this configuration, for example, when the angle between the chassis is set to about 120 degrees and the electronic apparatus is used as a laptop PC, a lower edge portion of the second chassis is positioned above the first chassis. Therefore, in this configuration, the lower bezel of the display provided in the second chassis is large and exposed to the exterior, and the appearance quality of the entire apparatus is deteriorated.

[0006] In order to solve this problem, it is also considered to reverse the rotation timing of each shaft. Specifically, it is also considered that only the rotation shaft on the first chassis side having the keyboard is rotated from 0 degrees to 180 degrees, and only the rotation shaft on the second chassis side having the display is rotated from 180 degrees to 360 degrees. However, in this case, the lower edge portion of the second chassis is disposed below the first chassis at an angle greater than 90 degrees, for example, at about 120 degrees for use as a laptop PC. As a result, in this case, the lower edge portion of the second chassis comes into contact with a desk surface or the like, causing problems of rattling and damage. In addition, when the lower edge portion of the second chassis comes into contact with a desk surface or the like, the first chassis having the keyboard will lift up, and the two chassis will not be disposed flat when viewed from the side at a position of 180 degrees.

[0007] Embodiments of the present disclosure provide an electronic apparatus and a hinge device that improves the appearance quality and suppress rattling and damage to the chassis during use.

[0008] An electronic apparatus according to a first aspect of the present invention includes a first chassis having a first surface and a second surface, a second chassis having a first surface and a second surface and adjacent to the first chassis, and a hinge device configured to connect the first

chassis and the second chassis to be rotatable from a first folded form in which first surfaces of the first chassis and the second chassis face each other to a second folded form in which second surfaces of the first chassis and the second chassis face each other, by connecting edge portions of the first chassis and the second chassis adjacent to each other to be rotatable relative to each other, in which the hinge device includes a first shaft fixed relative to the first chassis, a second shaft fixed relative to the second chassis, a support member configured to support the first shaft and the second shaft so as to be rotatable relative to each other with axial directions parallel to each other, a first slider having a first end portion facing a first shaft side and a second end portion facing a second shaft side, and configured to be slidable between the first shaft and the second shaft, a second slider having a first end portion facing the first shaft side and a second end portion facing the second shaft side and configured to be slidable between the first shaft and the second shaft, a first cam member fixed to the first shaft and having an outer peripheral surface provided with a locking part to which the first end portion of the second slider is fittable, and a second cam member fixed to the second shaft and having an outer peripheral surface provided with a first recessed portion into which the second end portion of the first slider is fittable and a second recessed portion into which the second end portion of the second slider is fittable.

[0009] A hinge device according to a second aspect of the present invention is a hinge device that connects a first chassis and a second chassis of an electronic apparatus to be rotatable, the hinge device including a first shaft fixed relative to the first chassis, a second shaft fixed relative to the second chassis, a support member configured to support the first shaft and the second shaft so as to be rotatable relative to each other with axial directions parallel to each other, a first slider having a first end portion facing a first shaft side and a second end portion facing a second shaft side, and configured to be slidable between the first shaft and the second shaft, a second slider having a first end portion facing the first shaft side and a second end portion facing the second shaft side and configured to be slidable between the first shaft and the second shaft, a first cam member fixed to the first shaft and having an outer peripheral surface provided with a locking part to which the first end portion of the second slider is fittable, and a second cam member fixed to the second shaft and having an outer peripheral surface provided with a first recessed portion into which the second end portion of the first slider is fittable and a second recessed portion into which the second end portion of the second slider is fittable.

[0010] According to the above-described aspect of the present invention, it is possible to improve the appearance quality and to suppress rattling and damage to the chassis during use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a schematic plan view of an electronic apparatus according to one or more embodiments as viewed from above.

[0012] FIG. 2 is a schematic side view of a hinge device and a peripheral portion thereof when a first chassis and a second chassis are in a first folded form.

[0013] FIG. 3 is a view in which the first chassis and the second chassis illustrated in FIG. 2 are at a first angular position.

[0014] FIG. 4 is a view in which the first chassis and the second chassis illustrated in FIG. 3 are at a position of 120 degrees.

[0015] FIG. 5 is a view in which the first chassis and the second chassis illustrated in FIG. 4 are at a position of 180 degrees.

[0016] FIG. 6 is a view in which the first chassis and the second chassis illustrated in FIG. 5 are at a second angular position.

[0017] FIG. 7 is a view in which the first chassis and the second chassis illustrated in FIG. 6 are in

a second folded form.

[0018] FIG. **8** is a perspective view of the hinge device.

[0019] FIG. **9** is a front view of the hinge device.

[0020] FIGS. **10A-10C** are side cross-sectional views of the hinge device in the first folded form, in which FIG. **10A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **10B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **10C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0021] FIGS. **11A-11C** are side cross-sectional views of the hinge device at a position of 60 degrees, in which FIG. **11A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **11B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **11C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0022] FIGS. **12A-12C** are side cross-sectional views of the hinge device at the first angular position, in which FIG. **12A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **12B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **12C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0023] FIGS. **13A-13C** are side cross-sectional views of the hinge device at a position of 120 degrees, in which FIG. **13A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **13B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **13C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0024] FIGS. **14A-14C** are side cross-sectional views of the hinge device at a position of 180 degrees, in which FIG. **14A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **14B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **14C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0025] FIGS. **15A-15C** are side cross-sectional views of the hinge device at the second angular position, in which FIG. **15A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **15B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **15C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0026] FIGS. **16A-16C** are side cross-sectional views of the hinge device at a position of 300 degrees, in which FIG. **16A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **16B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **16C** illustrates a cross section taken along the line C-C in FIG. **9**.

[0027] FIGS. **17A-17C** are side cross-sectional views of the hinge device at a position of 360 degrees, in which FIG. **17A** illustrates a cross section taken along the line A-A in FIG. **9**, FIG. **17B** illustrates a cross section taken along the line B-B in FIG. **9**, and FIG. **17C** illustrates a cross section taken along the line C-C in FIG. **9**.

DETAILED DESCRIPTION OF THE INVENTION

[0028] Embodiments of an electronic apparatus and a hinge device according to the present invention will be described in detail below with reference to the accompanying drawings.

[0029] FIG. **1** is a schematic plan view of an electronic apparatus **10** according to one or more embodiments as viewed from above. The electronic apparatus **10** has a configuration in which a first chassis **11** and a second chassis **12** are connected by left and right hinge devices **14L** and **14R** so as to be rotatable relative to each other. Hereinafter, the hinge devices **14L** and **14R** may be collectively referred to as a “hinge device **14**”.

[0030] The electronic apparatus **10** is a so-called convertible type PC that can be used as a laptop PC (laptop mode) or a tablet type PC (tablet mode) depending on an angular position between the chassis **11** and **12**. The laptop mode is a state in which the second chassis **12** is rotated with respect to the first chassis **11** at, for example, an angular position of about 120 degrees. The tablet mode is a state in which the second chassis **12** is rotated with respect to the first chassis **11** at, for example, an angular position of 360 degrees. The electronic apparatus **10** may be a mobile phone, a smartphone, an electronic notebook, a game machine, or the like.

[0031] The hinge device **14** can be moved down with an edge portion **11c** of the hinge device **14** side of the second chassis **12** protruding backward during an opening operation. Accordingly, the laptop mode is a so-called drop-down structure in which a part or all of a lower bezel **11d** of a display **20** is hidden behind the first chassis **11** (see FIGS. **3** and **4**).

[0032] FIGS. **2** to **7** are schematic side views of the hinge device **14** and the peripheral portion thereof at each angular position when the second chassis **12** of the electronic apparatus **10** illustrated in FIG. **1** is rotated from 0 degrees to 360 degrees with respect to the first chassis **11**.

[0033] Hereinafter, a direction viewed from an operator who uses the electronic apparatus **10** in a laptop mode as illustrated in FIG. **4** will be referred to as a reference, and a depth direction will be referred to as front and rear, a width direction will be referred to as left and right, and a thickness direction will be referred to as up and down.

[0034] For convenience of description, the angular position of the first chassis **11** and the second chassis **12** is referred to as a 0-degree position (see FIG. **2**), in which the second chassis **12** is closed with respect to the first chassis **11** and a front surface (first surface) **12a** of the second chassis **12** faces an upper surface (first surface) **11a** of the first chassis **11**. The following description is based on the 0-degree position, with the angle ticking in the direction of opening while rotating the second chassis **12**. For example, a posture in which the second chassis **12** and the first chassis **11** are substantially orthogonal to each other is referred to as a 90-degree position. A posture in which the front surface **12a** of the second chassis **12** and the upper surface **11a** of the first chassis **11** face the same direction (upward) and are parallel to each other is referred to as a 180-degree position (see FIG. **5**). A posture in which a rear surface (second surface) **12b** of the second chassis **12** and a bottom surface (second surface) **11b** of the first chassis **11** face each other is referred to as a 360-degree position (see FIG. **7**). The 0-degree position, 90-degree position, 120-degree position, 180-degree position, 360-degree position, and the like can naturally deviate slightly from the exact angular positions indicated by the angle numbers, depending on the structure of the chassis **11**, **12**, or the hinge device **14**. For convenience, these deviated angular positions will be referred to as a 0-degree position or the like.

[0035] As illustrated in FIGS. **1** to **7**, in the electronic apparatus **10**, the edge portion (one edge portion) **11c** on a rear side of the first chassis **11** and an edge portion (one edge portion) **12c** on a lower side of the second chassis **12** are adjacent to each other. The edge portions **11c** and **12c** are connected so as to be rotatable relative to each other by a pair of left and right hinge devices **14L** and **14R**.

[0036] The first chassis **11** is a flat box having a rectangular outer shape in a plan view. Inside the first chassis **11**, various electronic components such as a motherboard on which a CPU and the like are mounted, a battery device, a memory, an antenna device, and the like are accommodated. A keyboard **16** and a touch pad **17** face the upper surface **11a** of the first chassis **11**. The reference numeral **18** in FIG. **2** denotes a rubber foot that serves as a leg portion when the bottom surface **11b** is placed on a placement surface **19** such as a desk surface.

[0037] The second chassis **12** has a rectangular outer shape in a plan view and is a flat box that is thinner than the first chassis **11**. A display surface **20a** of the display **20** faces the front surface **12a** of the second chassis **12**. The display **20** can be configured with a liquid crystal display or an organic EL display. The display surface **20a** is able to display a video or an image. The front surface **12a** of the second chassis **12** is covered with a glass plate **21** on substantially the entire surface thereof including the display surface **20a** of the display **20**. The glass plate **21** can be configured with touch glass corresponding to a touch operation. The reference numeral **22** in FIG. **1** denotes a camera.

[0038] As illustrated in FIGS. **2** to **7**, the hinge device **14** is able to rotate the chassis **11** and **12** from a 0-degree position (first folded form) to a 360-degree position (second folded form) by a biaxial structure. FIG. **2** illustrates a 0-degree position, FIG. **3** illustrates a 100-degree position, FIG. **4** illustrates a 120-degree position, FIG. **5** illustrates a 180-degree position, FIG. **6** illustrates a

280-degree position, and FIG. 7 illustrates a 360-degree position.

[0039] The hinge devices **14L** and **14R** include a first shaft **24**, a second shaft **26**, and a hinge chassis **28**. The first shaft **24** is relatively fixed to the first chassis **11**. The second shaft **26** is relatively fixed to the second chassis **12**. That is, the first shaft **24** rotates integrally with the first chassis **11**. The second shaft **26** rotates integrally with the second chassis **12**. The hinge chassis **28** supports the shafts **24** and **26** so as to be rotatable relative to each other. The hinge chassis **28** can also have a bar shape that is long and extends over the left and right hinge devices **14L** and **14R**. [0040] The hinge chassis **28** is disposed substantially along the up-down direction at a 0-degree position (see FIG. 2) and a 360-degree position (see FIG. 7). At the 0-degree position, the second shaft **26** is disposed above the first shaft **24**. At the 360-degree position, the second shaft **26** is disposed below the first shaft **24**. Accordingly, in the electronic apparatus **10**, the front-rear direction positions of the edge portions **11c** and **12c** of the chassis **11** and **12** substantially coincide with each other at the 0-degree position and the 360-degree position. At the 0-degree position and the 360-degree position, the front-rear direction positions of the other edge portions (front edge portions) of the chassis **11** and **12** opposite to the edge portions **11c** and **12c** also substantially coincide with each other.

[0041] The hinge device **14** includes a rotation shaft selection mechanism portion **30** that selects one of the first shaft **24** and the second shaft **26** and rotates only the selected shaft. In the hinge device **14**, rotation of the first shaft **24** is allowed and rotation of the second shaft **26** is stopped between the 0-degree position and the 100-degree position (first angular position) (see FIGS. 2 and 3). In the hinge device **14**, the rotation of the second shaft **26** is allowed and the rotation of the first shaft **24** is stopped between the 100-degree position and the 280-degree position (second angular position) (see FIGS. 3 to 6). In the hinge device **14**, the rotation of the first shaft **24** is allowed and the rotation of the second shaft **26** is stopped between the 280-degree position and the 360-degree position (see FIGS. 6 and 7). As described above, in the hinge device **14**, the first shaft **24** or the second shaft **26** is selected at two switching angular positions (for example, the 100-degree position and the 280-degree position). Accordingly, in the hinge device **14**, only one selected shaft rotates as a rotation shaft between the chassis **11** and **12**, and the rotation of the other shaft is locked.

[0042] Next, a specific configuration example of the hinge device **14** will be described.

[0043] FIG. 8 is a perspective view of the hinge device **14L**. FIG. 9 is a front view of the hinge device **14L**. FIGS. 8 and 9 are views illustrating the internal structure of the hinge device **14L** with the hinge chassis **28** detached. FIGS. 10A, 10B, and 10C are side cross-sectional views of the hinge device **14L** at the 0-degree position. Similarly, FIGS. 11A, 11B, and 11C illustrate a 60-degree position, FIGS. 12A, 12B, and 12C illustrate a 100-degree position, FIGS. 13A, 13B, and 13C illustrate a 120-degree position, FIGS. 14A, 14B, and 14C illustrate a 180-degree position, FIGS. 15A, 15B, and 15C illustrate a 280-degree position, FIGS. 16A, 16B, and 16C illustrate a 300-degree position, and FIGS. 16A, 16B, and 16C illustrate a 360-degree position. In FIG. 10, FIG. 10A illustrates a cross section taken along the line A-A in FIG. 9, FIG. 10B illustrates a cross section taken along the line B-B in FIG. 9, and FIG. 10C illustrates a cross section taken along the line C-C in FIG. 9. Similarly, FIGS. 11A, 11B, 11C, and the like in FIGS. 11A to 17C illustrate cross sections taken along the lines A-A, B-B, and C-C in FIG. 9, respectively, and illustrate a state in which the hinge device **14L** is operated from FIGS. 10A, 10B, and 10C.

[0044] In one or more embodiments, the right hinge device **14R** can have the same or similar configuration as the left hinge device **14L** except that the right hinge device **14R** has a left-right symmetrical structure with respect to the left hinge device **14L**. Therefore, hereinafter, the left hinge device **14L** will be typically described, and specific description and illustration of the right hinge device **14R** will be omitted.

[0045] As illustrated in FIGS. 8 to 10, the hinge device **14L** includes the first shaft **24** and the second shaft **26** whose axial directions are parallel to each other, the hinge chassis **28** (see FIG. 2) that supports each of the shafts **24** and **26** so as to be rotatable relative to each other, and the

rotation shaft selection mechanism portion **30** that selectively switches between rotation timings of each of the shafts **24** and **26**.

[0046] The first shaft **24** and the second shaft **26** are cylindrical shafts. The axial directions of the respective shafts **24** and **26** are disposed in the directions along the longitudinal direction (the left-right direction) of the edge portions **11c** and **12c**. On the outer peripheral surfaces of the respective shafts **24** and **26**, cut surfaces for preventing rotation of the members (cam members **38** and **39** and the like) to be externally fitted and fixed are formed at appropriate positions.

[0047] A first bracket **32** is connected to one end (left end) of the first shaft **24** such that the first bracket **32** and the first shaft **24** are not rotatable relative to each other. The first bracket **32** is a metal plate for fixing the first shaft **24** to the first chassis **11** so that the first shaft **24** is not rotatable. The first bracket **32** is screwed to the inner surface of the first chassis **11**. A first torque generation unit **33** is attached to the other end (right end) of the first shaft **24**. The first torque generation unit **33** has a configuration in which, for example, a plurality of dish-shaped leaf springs **33a** are laminated in a row in the axial direction of the first shaft **24** and are pressed by nuts **33b**. The first torque generation unit **33** is able to apply a predetermined rotational torque to the rotation of the first shaft **24**.

[0048] A second bracket **34** is connected to one end (left end) of the second shaft **26** such that the second bracket **34** and the second shaft **26** are not rotatable relative to each other. The second bracket **34** is a metal plate for fixing the second shaft **26** to the second chassis **12** so that the second shaft **26** is not rotatable. The second bracket **34** is screwed to the inner surface of the second chassis **12**. A second torque generation unit **35** is attached to the other end (right end) of the second shaft **26**. The second torque generation unit **35** has a configuration in which, for example, a plurality of dish-shaped leaf springs **35a** are laminated in a row in the axial direction of the second shaft **26** and are pressed by a nut **35b**. The second torque generation unit **35** is able to apply a predetermined rotational torque to the rotation of the second shaft **26**.

[0049] As illustrated in FIGS. **8** to **10**, the rotation shaft selection mechanism portion **30** can include support plates **36** and **37**, a first cam member **38**, and a stopper cam member **40** in order from one end side to the other end side of the first shaft **24**. Hereinafter, the rotation shaft selection mechanism portion **30** may be simply referred to as a “mechanism portion **30**”. The mechanism portion **30** can include the support plates **36** and **37**, a second cam member **39**, and the stopper cam member **40** in order from one end side to the other end side of the second shaft **26**. The mechanism portion **30** can include the support plates **36** and **37**, a first slider **42**, a second slider **43**, and the stopper cam member **40** in order from one end side to the other end side in a space between the first shaft **24** and the second shaft **26** (inter-shaft space).

[0050] The hinge chassis **28** is provided to cover each constituent element of the shafts **24** and **26** and the mechanism portion **30**. The hinge chassis **28** can be rotated around the shaft of the first shaft **24** or the second shaft **26** selected by the mechanism portion **30** (see FIGS. **2** to **7**). During the rotation, the hinge chassis **28** is integrally rotated with the support plates **36** and **37**, the stopper cam member **40**, and the like.

[0051] The support plates **36** and **37** are substantially flat elliptical metal plates that are bridged over the first shaft **24** and the second shaft **26**. The support plates **36** and **37** have shaft holes into which the respective shafts **24** and **26** are inserted so as to be rotatable relative to each other at both end portions. Accordingly, the support plates **36** and **37** function as a support member that support each of the shafts **24** and **26** so as to be rotatable relative to each other while maintaining the inter-shaft distance between the shafts **24** and **26**. In the configuration example illustrated in FIGS. **8** and **9**, the two support plates **36** and **37** are arranged in parallel as a set of two, but only one support plate **36** or **37** may be used.

[0052] The first cam member **38** is a cylindrical metal member. The first cam member **38** is externally fitted and fixed in a state in which the first cam member **38** is not rotatable relative to the first shaft **24**. A first arc-shaped groove portion **38b** and a second arc-shaped groove portion **38c**

can be formed on an outer peripheral surface **38a** of the first cam member **38**. The arc-shaped groove portions **38b** and **38c** are arranged in the axial direction of the first shaft **24**. The first arc-shaped groove portion **38b** and the second arc-shaped groove portion **38c** can be misaligned by, for example, 100 degrees with respect to the circumferential direction of the outer peripheral surface **38a**. The 100 degrees is an angular difference between the 0-degree position and the first angular position (100-degree position). More specifically, the first arc-shaped groove portion **38b** may be misaligned by 100 degrees in the opening direction (direction from 0 degrees to 360 degrees) with respect to the second arc-shaped groove portion **38c**. In other words, the second arc-shaped groove portion **38c** can be misaligned by 100 degrees in the closing direction (direction from 360 degrees to 0 degrees) with respect to the first arc-shaped groove portion **38b**.

[0053] The first arc-shaped groove portion **38b** is located at a position at which a first end portion **42a** of the first slider **42** can face the first arc-shaped groove portion **38b** in the axial direction of the first shaft **24**. The second arc-shaped groove portion **38c** is located at a position at which a first end portion **43a** of the second slider **43** can face the second arc-shaped groove portion **38c** in the axial direction of the first shaft **24**. The first end portions **42a** and **43a** of the sliders **42** and **43** can be inserted into the arc-shaped groove portions **38b** and **38c**, respectively. The inserted first end portions **42a** and **43a** are movable relative to the arc-shaped groove portions **38b** and **38c** in the circumferential direction. The second arc-shaped groove portion **38c** has a locking part **38cl** which is a wall surface on one side in the circumferential direction of the second arc-shaped groove portion **38c**. The locking part **38cl** is able to lock the first end portion **43a** of the second slider **43** in a case in which the second shaft **26** is rotated from the 0-degree position to the first angular position (100-degree position) in the opening direction (see FIG. 12).

[0054] The second cam member **39** is a cylindrical metal member. The second cam member **39** is externally fitted and fixed to the second shaft **26** in a state in which the second cam member **39** and the second shaft **26** are not rotatable relative to each other. A first recessed portion **39b**, a second recessed portion **39c**, and a stopper groove portion **39d** can be formed on an outer peripheral surface **39a** of the second cam member **39**. The recessed portions **39b** and **39c** and the stopper groove portion **39d** are arranged in order in the axial direction of the second shaft **26**. The recessed portions **39b** and **39c** may be misaligned by, for example, 180 degrees with respect to the circumferential direction of the outer peripheral surface **39a**.

[0055] The first recessed portion **39b** is located at a position at which the second end portion **42b** of the first slider **42** can face the first recessed portion **39b** in the axial direction of the second shaft **26**. The second recessed portion **39c** is located at a position at which the second end portion **43b** of the second slider **43** can face the second recessed portion **39c** in the axial direction of the second shaft **26**. The second end portions **42b** and **43b** of the sliders **42** and **43** can be fitted into the recessed portions **39b** and **39c**, respectively. The recessed portions **39b** and **39c** is able to stop the relative movement of the fitted second end portions **42b** and **43b** in the circumferential direction. The recessed portions **39b** and **39c** can have, for example, a substantially arc-shaped cross section to which the second end portions **42b** and **43b** having an arc shape can be fitted and from which the second end portions **42b** and **43b** can be removed.

[0056] The stopper groove portion **39d** is located at a position facing the stopper portion **40c** of the stopper cam member **40** in the axial direction of the second shaft **26**. The stopper portion **40c** is inserted into the stopper groove portion **39d** and is movable relative to the stopper groove portion **39d** in the circumferential direction. The stopper groove portion **39d** has stopper surfaces **39d1** and **39d2** which are wall surfaces on one side and the other side of the stopper groove portion **39d** in the circumferential direction. The stopper surface (open stopper surface) **39d1** abuts against the stopper portion **40c** when the second shaft **26** is rotated from the first angular position (100-degree position) to the second angular position (280-degree position) in the opening direction. The stopper surface (close stopper surface) **39d2** abuts against the stopper portion **40c** when the second shaft **26** is rotated from the second angular position (280-degree position) to the first angular position (100-

degree position) in the closing direction.

[0057] The stopper cam member **40** can include a support plate **40a**, a pair of guide plates **40b** and **40b**, a stopper portion **40c**, and a protruding portion **40d**.

[0058] The support plate **40a** is a substantially flat elliptical metal plate that is bridged over the first shaft **24** and the second shaft **26**. The support plate **40a** has shaft holes into which the respective shafts **24** and **26** are inserted so as to be rotatable relative to each other at both end portions.

Accordingly, the support plate **40a** functions as a support member that supports each of the shafts **24** and **26** so as to be rotatable relative to each other while maintaining the inter-shaft distance between the shafts **24** and **26**. The support plate **40a** is held to sandwich the sliders **42** and **43** between the support plate **40a** and the support plate **37**. Accordingly, the sliders **42** and **43** are prevented from moving in the axial directions of the shafts **24** and **26**.

[0059] The pair of guide plates **40b** and **40b** are protruding pieces that protrude from the support plate **40a** toward the support plate **37** side. The pair of guide plates **40b** and **40b** can be disposed in a line-symmetrical manner with respect to a direction connecting the shaft centers of the shafts **24** and **26** (hereinafter, also referred to as an “inter-shaft direction D”). The pair of guide plates **40b** and **40b** is able to guide the sliders **42** and **43** to be slidable along the inter-shaft direction D between the facing surfaces thereof.

[0060] A protrusion **40b1** is formed on an end surface of each guide plate **40b** on the support plate **37** side. The protrusion **40b1** is fitted into a groove formed in the outer peripheral edge surface of the support plate **37**. As a result, each guide plate **40b** is prevented from being misaligned in the inter-shaft direction D. Further, each guide plate **40b** is also prevented from being misaligned in a direction orthogonal to the surface normal direction.

[0061] The stopper cam member **40** has a thick-walled portion **40e** in which a side surface of the support plate **40a** on the guide plate **40b** side is bulged. The thick-walled portion **40e** enters between the shafts **24** and **26**. The stopper portion **40c** is a protrusion that protrudes from the upper surface of the thick-walled portion **40e**. The stopper portion **40c** is inserted into the stopper groove portion **39d** of the second cam member **39**. When the second cam member **39** is rotated together with the second shaft **26**, the stopper portion **40c** is able to stop further rotation of the second shaft **26** by abutting against the stopper surfaces **39d1** and **39d2** of the stopper groove portion **39d**.

[0062] The protruding portion **40d** protrudes from a side surface of the support plate **40a** opposite to the thick-walled portion **40e** side and is disposed between the torque generation units **33** and **35**. A female screw that tightens the screw for fastening the hinge chassis **28** is formed in the protruding portion **40d**.

[0063] The first slider **42** and the second slider **43** are metal members having a substantially flat elliptical cross-sectional shape. The first slider **42** has the first end portion **42a** facing the first shaft **24** side and the second end portion **42b** facing the second shaft **26** side. The second slider **43** has the first end portion **43a** facing the first shaft **24** side and the second end portion **43b** facing the second shaft **26** side. The end portions **42a**, **42b**, **43a**, and **43b** can have an arc shape.

[0064] The sliders **42** and **43** are each independently slidable along the inter-shaft direction D. The first slider **42** is disposed at a position at which the first slider **42** can face the first arc-shaped groove portion **38b** of the first cam member **38** and the first recessed portion **39b** of the second cam member **39** with respect to the axial directions of the shafts **24** and **26**. The second slider **43** is disposed at a position at which the second slider **43** can face the second arc-shaped groove portion **38c** of the first cam member **38** and the second recessed portion **39c** of the second cam member **39**.

[0065] Rectangular hollow portions **42c** and **43c** extending in the inter-shaft direction D are formed at the centers of the sliders **42** and **43**, respectively. A guide rod **46** extending along the axial direction between the shafts **24** and **26** is inserted into the hollow portions **42c** and **43c**. One end portion of the guide rod **46** in the longitudinal direction is fitted and supported by the support plates **36** and **37**, and the other end portion of the guide rod **46** is fitted and supported by the thick-walled portion **40e**. The width (height) of the guide rod **46** in the inter-shaft direction D is shorter than the

width (height) of the hollow portions **42c** and **43c** in the inter-shaft direction **D**. The outer peripheral side surfaces of the sliders **42** and **43** are guided by the pair of guide plates **40b** and **40b**, and further the hollow portions **42c** and **43c** are guided by the guide rod **46**. Accordingly, the sliders **42** and **43** can smoothly slide along the inter-shaft direction **D**.

[0066] Next, a rotational operation of the chassis **11** and **12** by the hinge device **14** will be described.

[0067] The operation and action of the hinge device **14** when the second chassis **12** at the 0-degree position is opened to the 360-degree position will be described. This operation is performed, for example, by a user gripping the upper end portion (end portion on a side opposite to the edge portion **12c** side) of the second chassis **12** and applying an external force in the opening direction.

[0068] An operation from the 0-degree position to the first angular position (100-degree position) will be described.

[0069] In this case, the first end portion **42a** of the first slider **42** is disposed at a position at which the first end portion **42a** can slide on the outer peripheral surface **38a** of the first cam member **38** (see FIG. **10A**). The first end portion **43a** of the second slider **43** is disposed at a position at which the second slider **43** is movable relative to the second arc-shaped groove portion **38c** of the first cam member **38** in the circumferential direction (see FIG. **10B**). Therefore, the rotation of the first shaft **24** is allowed. On the other hand, the second end portion **42b** of the first slider **42** is fitted to the first recessed portion **39b** of the second cam member **39** (see FIG. **10A**). That is, the first slider **42** is in a state in which the first slider **42** cannot slide in the inter-shaft direction **D**. Therefore, the rotation of the second shaft **26** is stopped.

[0070] Between the 0-degree position and the first angular position, the stopper portion **40c** of the stopper cam member **40** abuts against the stopper surface **39d2** of the second cam member **39** (see FIG. **10C**). The stopper portion **40c** is located at a position at which the stopper portion **40c** is movable relative to the stopper groove portion **39d** in the opening direction.

[0071] Therefore, as illustrated in FIGS. **10** to **12**, between the 0-degree position and the first angular position, the first shaft **24** is a rotation shaft and the second shaft **26** is locked. Therefore, the second chassis **12** rotates with the first shaft **24** as a rotation shaft. In this case, the hinge chassis **28** also rotates with the first shaft **24** as a rotation shaft (see FIGS. **2** and **3**).

[0072] Accordingly, at the first angular position at which using in the laptop mode is expected, the shaft center of the second shaft **26** is disposed below the shaft center of the first shaft **24** in the thickness direction (up-down direction) of the first chassis **11** (see FIGS. **3** and **12**). That is, the edge portion **12c** of the second chassis **12** is disposed below the upper surface **11a** of the first chassis **11** in the up-down direction (drop down). As a result, the electronic apparatus **10** is able to hide a part or all of the lower bezel **11d** of the display **20** behind the first chassis **11**, and the appearance quality is improved. In this case, the edge portion **12c** of the second chassis **12** does not come into contact with the placement surface **19** (see FIG. **3**). In other words, the electronic apparatus **10** is able to employ a rubber foot **18** that secures a height at which the edge portion **12c** does not come into contact with the placement surface **19** at the first angular position illustrated in FIG. **3**.

[0073] By the way, in the first folded form (0-degree position), the stopper portion **40c** is at a position at which the stopper portion **40c** cannot move relative to the stopper groove portion **39d** in the closing direction (see FIG. **10C**). That is, in the first folded form of the electronic apparatus **10**, the first slider **42** is fitted into the first recessed portion **39b**, and at the same time, the stopper portion **40c** is locked by the stopper surface **39d2**. Accordingly, the rotation of the second chassis **12** in the closing direction is more reliably stopped. Therefore, in the first folded form in which it is expected that the electronic apparatus **10** will be carried around, the positional relationship between the chassis **11** and **12** is more firmly maintained. Therefore, in the electronic apparatus **10**, rattling between the chassis **11** and **12** is suppressed during carrying or the like.

[0074] An operation from the first angular position (100-degree position) to the 120-degree

position will be described.

[0075] At the first angular position illustrated in FIG. 12, the first end portion **42a** of the first slider **42** is disposed at a position at which the first end portion **42a** can be inserted into the first arc-shaped groove portion **38b** of the first cam member **38** (see FIG. 12A). That is, the first slider **42** is in a state in which it can slide in the inter-shaft direction D. On the other hand, the first end portion **43a** of the second slider **43** is locked to the locking part **38cl** of the second arc-shaped groove portion **38c** (see FIG. 12B). The second end portion **43b** is disposed at a position at which the second end portion **43b** can slide on the outer peripheral surface **39a** of the second cam member **39**. That is, the second slider **43** is in a state in which the second slider **43** cannot slide in the inter-shaft direction D. Therefore, the rotation of the first shaft **24** is stopped. In other words, any further rotation of the second chassis **12** (hinge chassis **28**) about the first shaft **24** is stopped.

[0076] When the second chassis **12** receives a rotational force toward the 120-degree position from the first angular position, the second shaft **26** receives the rotational force. In this case, the first recessed portion **39b** of the second cam member **39** presses the second end portion **42b** of the first slider **42** with the edge wall thereof. Accordingly, the first slider **42** slides to the first shaft **24** side. The first end portion **42a** is inserted into the first arc-shaped groove portion **38b** (see FIG. 13A). In this case, the second end portion **42b** of the first slider **42** is disposed at a position at which the second cam member **39** can slide on the outer peripheral surface **39a**. Therefore, the rotation of the second shaft **26** is allowed. The stopper portion **40c** moves relative to the stopper groove portion **39d** between the first angular position and the 120-degree position.

[0077] Therefore, as illustrated in FIGS. 12 and 13, between the first angular position and the 120-degree position, the second shaft **26** is a rotation shaft and the first shaft **24** is locked. Therefore, the second chassis **12** rotates with the second shaft **26** as a rotation shaft. In this case, the hinge chassis **28** does not rotate (see FIGS. 3 and 4).

[0078] Accordingly, even at the 120-degree position at which using in the laptop mode is expected, the height relationship between the second shaft **26** and the first shaft **24** is maintained at the first angular position. That is, the shaft center of the second shaft **26** is below the shaft center of the first shaft **24** (see FIGS. 4 and 13). In general, the electronic apparatus **10** is most suitable for the angle range between the 100-degree position and the 120-degree position in the laptop mode. In the electronic apparatus **10**, in this angle range, a state in which the edge portion **12c** is below the upper surface **11a** is maintained. That is, the electronic apparatus **10** maintains the drop-down state at least within this angle range, and it is possible to obtain high appearance. In addition, between the first angular position and the 120-degree position, the edge portion **12c** of the second chassis **12** does not move in the up-down direction (see FIGS. 3, 4, 12 and 13). Therefore, the contact of the edge portion **12c** with the placement surface **19** is suppressed within the above-described angle range in which the electronic apparatus **10** is expected to be used in the laptop mode. As a result, the electronic apparatus **10** is prevented from rattling on the placement surface **19**, and the second chassis **12** is prevented from being damaged on the placement surface **19**.

[0079] An operation from the 120-degree position to the second angular position (280-degree position) will be described.

[0080] The operation in this case is the same as the operation from the first angular position to the 120-degree position (see FIGS. 13 to 15). That is, the first end portion **43a** of the second slider **43** is locked by the locking part **38cl** of the first cam member **38**. Therefore, the rotation of the first shaft **24** is stopped. On the other hand, the second end portions **42b** and **43b** of the sliders **42** and **43** are disposed at positions at which the outer peripheral surface **39a** of the second cam member **39** can slide. Therefore, the rotation of the second shaft **26** is allowed. The stopper portion **40c** moves relative to the stopper groove portion **39d** between the 120-degree position and the second angular position.

[0081] An operation from the second angular position (280-degree position) to the 360-degree position will be described.

[0082] At the second angular position illustrated in FIG. 15, the second recessed portion **39c** of the second cam member **39** is disposed at a position facing the second end portion **43b** of the second slider **43** (see FIG. 15B). That is, the second slider **43** is in a state of being slidable in the inter-shaft direction D. On the other hand, the first end portion **42a** of the first slider **42** is disposed at a position at which the first end portion **42a** is movable relative to the arc-shaped groove portion **38b** of the first cam member **38** (see FIG. 15A). The second end portion **42b** is disposed at a position at which the second end portion **42b** can slide on the outer peripheral surface **39a** of the second cam member **39**.

[0083] When the second chassis **12** receives a rotational force from the second angular position to the 360-degree position, the first shaft **24** receives the rotational force. In this case, the locking part **38cl** of the first cam member **38** presses the first end portion **43a** of the second slider **43**. As a result, the second slider **43** slides to the second shaft **26** side. The second end portion **43b** is fitted into the second recessed portion **39c** (see FIG. 16B). In this case, the first end portion **43a** of the second slider **43** is separated from the second arc-shaped groove portion **38c** and is disposed at a position at which the first cam member **38** can slide on the outer peripheral surface **38a**. Therefore, the rotation of the second shaft **26** is stopped, and the rotation of the first shaft **24** is allowed.

[0084] At the second angular position, the stopper portion **40c** abuts against the stopper surface **39d1** of the stopper groove portion **39d**. Therefore, further rotation of the second shaft **26** is more reliably stopped. Therefore, when the second slider **43** is rotated in the opening direction from the second angular position, the second slider **43** slides more smoothly.

[0085] Accordingly, as illustrated in FIGS. 15 to 17, between the second angular position and the 360-degree position, the first shaft **24** is a rotation shaft again, and the second shaft **26** is locked. Therefore, the second chassis **12** rotates with the first shaft **24** as a rotation shaft. In this case, the hinge chassis **28** is also rotated with the first shaft **24** as a rotation shaft (see FIGS. 6 and 7). Accordingly, the electronic apparatus **10** can be set in the second folded form (360-degree position) and can be used in the tablet mode.

[0086] The stopper portion **40c** abuts against the stopper surface **39d1** between the second angular position and the 360-degree position. That is, the stopper portion **40c** is located at a position at which the stopper portion **40c** is movable relative to the stopper groove portion **39d** in the closing direction.

[0087] On the other hand, in the second folded form (360-degree position), the stopper portion **40c** is at a position at which the stopper portion **40c** cannot move relative to the stopper groove portion **39d** in the opening direction. That is, in the second folded form of the electronic apparatus **10**, the second slider **43** is fitted into the second recessed portion **39c**, and at the same time, the stopper portion **40c** is locked by the stopper surface **39d1**. In the electronic apparatus **10**, the rotation of the second chassis **12** in the opening direction in the second folded form is more reliably stopped. Therefore, in the second folded form in which the electronic apparatus **10** is expected to be used in the tablet mode, the positional relationship between the chassis **11** and **12** is more firmly maintained. In the electronic apparatus **10**, for example, rattling when a touch operation is performed on the display **20** and shaking of the second chassis **12** are suppressed.

[0088] The operation of the hinge device **14** when the second chassis **12** at the 360-degree position is closed to the 0-degree position is an operation opposite to the opening operation described above, and thus, the detailed description thereof will be omitted. That is, the hinge device **14** operates as illustrated in FIGS. 17 to 10.

[0089] As described above, the electronic apparatus **10** and the hinge device **14** can include the first slider **42** that is slidable between the first shaft **24** and the second shaft **26** and the second slider **43** that is slidable between the first shaft **24** and the second shaft **26**. The electronic apparatus **10** and the hinge device **14** can further include the first cam member **38** that is fixed to the first shaft **24** and is provided with the locking part **38cl** to which the first end portion **43a** of the second slider **43** is fittable on the outer peripheral surface **38a**, and the second cam member **39** that is fixed to the

second shaft **26** and is provided with the first recessed portion **39b** into which the second end portion **42b** of the first slider **42** can be fitted and the second recessed portion **39c** into which the second end portion **43b** of the second slider **43** can be fitted, on the outer peripheral surface **39a**. [0090] Accordingly, the hinge device **14** is able to select the shafts **24** and **26** that act as the rotation shaft at a desired rotation timing by the stop action of the cam members **38** and **39** by the two sliders **42** and **43**. Therefore, the electronic apparatus **10** is able to easily switch between the shafts **24** and **26** as a rotation shaft at the first angular position on the closing side from the 120-degree position expected as most preferred for using in, for example, the laptop mode. Therefore, when the electronic apparatus **10** is used in the laptop mode, it is possible to prevent the second chassis **12** from coming into contact with the placement surface **19**, which may result in rattling or damage. In addition, when the electronic apparatus **10** is used in the laptop mode, the second chassis **12** is prevented from coming into contact with the placement surface **19** and the first chassis **11** is prevented from being lifted. Furthermore, since the electronic apparatus **10** has the two chassis **11** and **12** disposed substantially flat when viewed from the side at a 180-degree position, the appearance quality is further improved. Moreover, when the electronic apparatus **10** is used in the laptop mode, the edge portion **12c** of the second chassis **12** is disposed behind and below the first chassis **11**, and thus it is possible to obtain an even higher appearance quality.

[0091] In particular, the hinge device **14** includes two sliders **42** and **43** that slide between the shafts **24** and **26**, and two cam members **38** and **39** which the sliders **42** and **43** are fitted to and slide on. The hinge device **14** is configured to selectively switch between the rotation shaft at the first angular position and the second angular position. In other words, the hinge device **14** realizes the switching of the rotation shaft twice without requiring a member that moves in the axial directions of the shafts **24** and **26**. Therefore, the hinge device **14** is able to simplify the configuration and is also able to suppress the axial width of the entire device, and thus it is easy to reduce the size. Further, since the hinge device **14** does not have a member that moves in the axial directions of the shafts **24** and **26**, it is possible to suppress the hindrance to the rotation of each of the shafts **24** and **26**, and it is possible to perform a smoother rotational operation.

[0092] The hinge device **14** can include the stopper surface **39d1** that is formed on the outer peripheral surface **39a** of the second cam member **39** and faces the circumferential direction thereof, and the stopper portion **40c** that abuts against the stopper surface **39d1**. For example, the stopper portion **40c** can be configured to abut against the stopper surface **39d1** at the second angular position (280 degrees) in a case of the opening operation (see FIG. 15C). In this case, the sliding of the second slider **43** is smoother, and the switching from the second shaft **26** to the first shaft **24** is smoother.

[0093] The hinge device **14** can further include the stopper surface **39d2** that is formed on the outer peripheral surface **39a** of the second cam member **39** and that is provided on a side opposite to the stopper surface **39d1** facing the circumferential direction thereof. For example, the stopper portion **40c** can be configured to abut against the stopper surface **39d2** at the first angular position (100 degrees) in a case of the closing operation (see FIG. 12C). In this case, the sliding of the first slider **42** is smoother, and the switching from the second shaft **26** to the first shaft **24** is smoother.

[0094] The present invention is not limited to the above-described embodiments, and it goes without saying that the present invention can be freely modified without departing from the gist of the present invention.

DESCRIPTION OF SYMBOLS

[0095] **10** electronic apparatus [0096] **11** first chassis [0097] **12** second chassis [0098] **14L**, **14R** hinge device [0099] **16** keyboard [0100] **20** display [0101] **24** first shaft [0102] **26** second shaft [0103] **28** hinge chassis [0104] **30** rotation shaft selection mechanism portion [0105] **36**, **37**, **40a** support plate [0106] **38** first cam member [0107] **38a**, **39a** outer peripheral surface [0108] **38b** first arc-shaped groove portion [0109] **38c** second arc-shaped groove portion [0110] **39** second cam member [0111] **39b** first recessed portion [0112] **39c** second recessed portion [0113] **39d1**, **39d2**

stopper surface [0114] **40** stopper cam member [0115] **40c** stopper portion [0116] **42** first slider [0117] **43** second slider

Claims

1. An electronic apparatus comprising: a first chassis having a first surface and a second surface; a second chassis having a first surface and a second surface and adjacent to the first chassis; and a hinge device configured to connect the first chassis and the second chassis to be rotatable from a first folded form in which first surfaces of the first chassis and the second chassis face each other to a second folded form in which second surfaces of the first chassis and the second chassis face each other, by connecting edge portions of the first chassis and the second chassis adjacent to each other to be rotatable relative to each other, wherein the hinge device includes a first shaft fixed relative to the first chassis, a second shaft fixed relative to the second chassis, a support member configured to support the first shaft and the second shaft so as to be rotatable relative to each other with axial directions parallel to each other, a first slider having a first end portion facing a first shaft side and a second end portion facing a second shaft side, and configured to be slidable between the first shaft and the second shaft, a second slider having a first end portion facing the first shaft side and a second end portion facing the second shaft side and configured to be slidable between the first shaft and the second shaft, a first cam member fixed to the first shaft and having an outer peripheral surface provided with a locking part to which the first end portion of the second slider is fittable, and a second cam member fixed to the second shaft and having an outer peripheral surface provided with a first recessed portion into which the second end portion of the first slider is fittable and a second recessed portion into which the second end portion of the second slider is fittable.
2. The electronic apparatus according to claim 1, wherein the first cam member has a first arc-shaped groove portion that extends in a circumferential direction of the outer peripheral surface, with the first end portion of the first slider movable relative to the first arc-shaped groove portion along an extending direction, and a second arc-shaped groove portion that extends in the circumferential direction of the outer peripheral surface, with the first end portion of the second slider movable relative to the second arc-shaped groove portion along an extending direction, and of which a wall surface on one end side in the extending direction is the locking part.
3. The electronic apparatus according to claim 2, wherein the first chassis and the second chassis sequentially pass through a first angular position and a second angular position when rotating from the first folded form to the second folded form, in the hinge device, when the first chassis and the second chassis are between the first folded form and the first angular position, rotation of the first shaft is allowed by the first end portion of the first slider being disposed at a position to be slidable on the outer peripheral surface of the first cam member and the first end portion of the second slider being disposed at a position to be movable relative to the second arc-shaped groove portion, while rotation of the second shaft is stopped by the second end portion of the first slider being fitted into the first recessed portion, when the first chassis and the second chassis are between the first angular position and the second angular position, the rotation of the first shaft is stopped by the first end portion of the second slider being locked by the locking part, while the rotation of the second shaft is allowed by the second end portion of the first slider and the second end portion of the second slider being both disposed at positions to be slidable on the outer peripheral surface of the second cam member, and when the first chassis and the second chassis are between the first angular position and the second folded form, the rotation of the first shaft is allowed by the first end portion of the first slider being disposed at a position to be movable relative to the first arc-shaped groove portion and the first end portion of the second slider being disposed at a position to be slidable on the outer peripheral surface of the first cam member, while the rotation of the second shaft is stopped by the second end portion of the second slider being fitted into the second recessed portion.

4. The electronic apparatus according to claim 3, wherein when the first chassis and the second chassis are in the first folded form, the first shaft and the second shaft are disposed with an axis center of the second shaft above an axis center of the first shaft, with respect to a thickness direction of the first chassis, and when the first chassis and the second chassis are in the first angular position, the first chassis and the second chassis are disposed with the axis center of the second shaft lower than the axis center of the first shaft, with respect to the thickness direction of the first chassis.
5. The electronic apparatus according to claim 4, wherein the first chassis is mounted with a keyboard facing the first surface, and the second chassis is mounted with a display facing the first surface.
6. The electronic apparatus according to claim 4, wherein the second cam member has a stopper surface facing a circumferential direction on the outer peripheral surface, and the hinge device further includes a stopper portion that abuts against the stopper surface.
7. A hinge device that connects a first chassis and a second chassis of an electronic apparatus to be rotatable, the hinge device comprising: a first shaft fixed relative to the first chassis; a second shaft fixed relative to the second chassis; a support member configured to support the first shaft and the second shaft so as to be rotatable relative to each other with axial directions parallel to each other; a first slider having a first end portion facing a first shaft side and a second end portion facing a second shaft side, and configured to be slidable between the first shaft and the second shaft; a second slider having a first end portion facing the first shaft side and a second end portion facing the second shaft side and configured to be slidable between the first shaft and the second shaft; a first cam member fixed to the first shaft and having an outer peripheral surface provided with a locking part to which the first end portion of the second slider is fittable; and a second cam member fixed to the second shaft and having an outer peripheral surface provided with a first recessed portion into which the second end portion of the first slider is fittable and a second recessed portion into which the second end portion of the second slider is fittable.
8. The hinge device according to claim 7, wherein the first cam member has a first arc-shaped groove portion that extends in a circumferential direction of the outer peripheral surface, with the first end portion of the first slider movable relative to the first arc-shaped groove portion along an extending direction, and a second arc-shaped groove portion that extends in the circumferential direction of the outer peripheral surface, with the first end portion of the second slider movable relative to the second arc-shaped groove portion along an extending direction, and of which a wall surface on one end side in the extending direction is the locking part.
9. The hinge device according to claim 8, wherein a hinge device connects the first chassis and the second chassis to be rotatable from a first folded form in which first surfaces of the first chassis and the second chassis face each other to a second folded form in which second surfaces of the first chassis and the second chassis face each other, by connecting edge portions of the first chassis and the second chassis adjacent to each other to be rotatable relative to each other, the first chassis and the second chassis sequentially pass through a first angular position and a second angular position when rotated from the first folded form to the second folded form, in the hinge device, when the first chassis and the second chassis are between the first folded form and the first angular position, rotation of the first shaft is allowed by the first end portion of the first slider being disposed at a position to be slidable on the outer peripheral surface of the first cam member and the first end portion of the second slider being disposed at a position to be movable relative to the second arc-shaped groove portion, while rotation of the second shaft is stopped by the second end portion of the first slider being fitted into the first recessed portion, when the first chassis and the second chassis are between the first angular position and the second angular position, the rotation of the first shaft is stopped by the first end portion of the second slider being locked by the locking part, while the rotation of the second shaft is allowed by the second end portion of the first slider and the second end portion of the second slider being both disposed at positions to be slidable on the outer

peripheral surface of the second cam member, and when the first chassis and the second chassis are between the first angular position and the second folded form, the rotation of the first shaft is allowed by the first end portion of the first slider being disposed at a position to be movable relative to the first arc-shaped groove portion and the first end portion of the second slider being disposed at a position to be slidable on the outer peripheral surface of the first cam member, while the rotation of the second shaft is stopped by the second end portion of the second slider being fitted into the second recessed portion.
